

Data / evaluation space

$$s = [x, R, v_b, u_a, u_c, v_c^n, z_{\text{ref}}]$$

 v_b

pose kinematics, rollout metrics

 $\mathcal{T}_{d \rightarrow m}$

$$v_r = v_b - \Delta_c$$

 $\mathcal{T}_{m \rightarrow d}$

$$v_b = v_r + \Delta_c$$

$$\Delta_c(R, v_c^n) = [R^\top v_c^n; 0]$$



ODE model space

$$y = [x, R, v_r, u_a, u_c, v_c^n, z_{\text{ref}}]$$

 v_r

hydrodynamic branch, velocity loss